ShelfWise — Product Requirements Document

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Product: ShelfWise
Document purpose: Shared product and implementation specification for the development team.

Document details

Field	Answer
Product name	ShelfWise
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GitHub repository	https://github.com/noluMolui/Shelfwise
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Change log

Version	Date	Sections changed	What changed	Regenerated?
v0.1	10 Nov 2026	All	First PRD draft based on the ShelfWise team proposal	None

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1. Product overview

1.1 Product name

ShelfWise

1.2 One-line pitch

ShelfWise helps small businesses manage inventory so that they can track stock, suppliers and purchasing needs in one place.

1.3 Problem statement

Many small businesses manage inventory using spreadsheets, handwritten records or basic tools that do not provide a complete view of their stock. These methods can make it difficult to know how much stock is available, identify products that are running low, track stock coming in and going out, and keep supplier information organised. The team proposal identifies five related problems: lost track of available stock, running out of popular products, poor stock management, difficulty tracking suppliers and limited visibility of inventory value. The proposal does not provide research statistics yet; the team must collect and add at least one credible number before approval.

1.4 Solution

ShelfWise gives a business one place to create and maintain product records, including SKU, category, price and quantity. Users can record incoming and outgoing stock, and the available quantity is updated when a stock movement is recorded. The platform identifies products that have reached their low-stock threshold and displays them as requiring attention. Users can maintain supplier records and associate suppliers with products. A dashboard summarises product counts, inventory value, low-stock products and recent stock activity.

1.5 Platform

Proposed: Web app that works in a browser. Design and build the responsive web experience first. Mobile-browser behaviour must be specified and tested before approval.

1.6 Vision

Within two years, ShelfWise could become a practical inventory platform for small businesses that need a clear, accessible way to manage products, stock movements and supplier information. Any expansion beyond the v1 scope will be considered after the MVP has been tested with real users.

1.7 Key terms

Term	Exact meaning in ShelfWise
Business	The small business whose inventory is managed in ShelfWise.
Product	An item that the business tracks in its inventory.
SKU	A business-defined unique stock-keeping code used to identify a product.
Stock movement	A recorded increase or decrease in a product’s stock quantity.
Supplier	A person or organisation that supplies products to the business.
Low-stock threshold	The minimum quantity set for a product; stock at or below this quantity is considered low.
Inventory value	The sum of each product’s recorded unit cost multiplied by its current quantity. **[TEAM DECISION REQUIRED: confirm which price field**

Term	Exact meaning in ShelfWise
	is used.]

Use these terms consistently in screens, code, database fields and team discussions.

2. Users, personas and roles

2.1 Primary user

Owners and staff members of small businesses who need to keep track of products and stock. The proposal does not define an industry, age range, location beyond the South African context implied by the team examples, device preference or connectivity profile. These details must be validated with target users.

2.2 Secondary users

- Business administrators or managers who oversee inventory and staff access.
- Staff members who add products or record stock movements.
- **[TEAM DECISION REQUIRED]** Whether suppliers receive accounts or use ShelfWise only through business users.

2.3 Personas

These are provisional personas for design discussion, not researched user profiles.

Field	Persona 1	Persona 2
Name and age	Thandi, thirty-five	Kabelo, twenty-seven
Role	Business Owner	Staff Member
Occupation / situation	Runs a small retail business	Works in the business and handles stock
Device and connectivity	Smartphone or laptop; connectivity to be validated	Smartphone or shared business computer; connectivity to be validated
Tech comfort	Medium — provisional	Medium — provisional
Language(s)	English; additional languages to be validated	English; additional languages to be validated
Goal	Understand current stock and what needs ordering	Record stock received and stock sold or removed
Frustrations today	Records may be spread across paper and spreadsheets	Unsure whether the recorded quantity is current
How the app helps	Provides a central inventory view and low-stock indicators	Provides a consistent process for recording stock movements
Quote	“I need to know what we have before I order more.”	“I need stock records that are easy to update.”

2.4 Roles and permissions

Proposed role model — confirm before implementation. The original proposal does not define account roles or multi-business access.

Role	How someone gets this role	Can do	Cannot do
Business Owner	Creates a business account or is assigned as its owner	Manage business products, suppliers, stock movements and business users; view dashboard	Access another business's records
Staff Member	Invited by the Business Owner **[invitation method TBD]**	View permitted inventory records; create products or stock movements according to permissions	Manage owner account or access another business's records
Platform Admin	**[TEAM DECISION REQUIRED]**	**[Permissions TBD]**	Must not access business data without an authorised support or operational reason

Every restriction must be enforced by the backend/database, not merely by hiding interface controls.

2.5 Admin

[PROPOSED — CONFIRM] In v1, platform administration will be handled through the Supabase dashboard, with no separate platform-admin screens. Business Owner functions remain available through the application.

3. Goals, success metrics and non-goals

3.1 Goals

Targets below are proposed MVP acceptance targets and should be confirmed by the team.

ID	Goal	Metric	Target	How to check
G1	Allow a business to create and maintain its product catalogue	Product create, update, search and delete tests passed	All defined product-management tests pass	Product records and acceptance tests
G2	Keep stock quantities accurate after recorded movements	Correct quantity after each stock movement	100% of valid stock-movement test cases pass	Stock movement records and automated tests
G3	Make low-stock	Correct products	100% of threshold	Product

ID	Goal	Metric	Target	How to check
	products visible	shown at or below their threshold	test cases pass	quantities, thresholds and dashboard
G4	Give owners a useful overview	Dashboard displays agreed summary measures	All agreed measures display correctly	Dashboard test checklist and 5–10 user tests

3.2 Non-goals

1. **Payments and checkout:** ShelfWise will not process customer payments in v1 because the core project is inventory management.
2. **Accounting and tax filing:** ShelfWise will not replace accounting software or submit tax returns.
3. **Automated supplier purchasing:** ShelfWise will not place orders with suppliers in v1; users will make purchasing decisions themselves.
4. **Barcode scanning:** Deferred to keep the MVP within scope.
5. **CSV import/export and bulk migration:** Deferred until the core product and stock workflows work reliably.
6. **Native Android or iOS apps:** v1 is a browser-based web app.

4. Features, business rules and notifications

4.1 Feature list

ID	Feature	What the user can do	Priority	Goal	Stories	Screens
F1	Product Management	Add, edit, delete, search and view products; maintain SKU, category, price and quantity-related product details	Must	G1	US1	S3, S4
F2	Stock Management	Record incoming and outgoing stock; view stock movement history; update current quantity	Must	G2	US2	S5, S6
F3	Low-Stock Alerts	Set a low-stock threshold and identify products at or below it	Must	G3	US3	S3, S7
F4	Supplier Management	Add, edit, view and delete supplier records; associate suppliers with products	Should	G1	US4	S8
F5	Business Dashboard	View total products, inventory value, low-stock products and recent stock	Must	G4	US5	S2

ID	Feature	What the user can do	Priority	Goal	Stories	Screens
		activity				

The five features reflect the proposal. The priority split is a proposed MVP interpretation: four Must features and one Should feature. The team should confirm whether Supplier Management is essential to the first release.

4.2 MVP definition

The ShelfWise MVP allows an authorised business user to sign in, maintain a product catalogue, record incoming and outgoing stock, identify products at or below a low-stock threshold, and view a dashboard of inventory information.

4.3 Future features

- CSV import and export.
- Product images.
- Barcode scanning.
- More detailed inventory reports.
- **[TEAM DECISION REQUIRED]** Any other future functionality.

4.4 Business rules

The following rules are proposed where the proposal gives enough direction. Items marked TBD must be decided before implementation.

ID	Rule	Used by
BR1	Every product belongs to exactly one business. A user may only access products belonging to a business they are authorised to access.	F1, F2, F3, F5
BR2	SKU must be unique within the same business. Two different businesses may use the same SKU.	F1
BR3	A product’s current stock quantity must not be changed by an ordinary product edit. Quantity changes must be recorded as stock movements. **[CONFIRM]**	F1, F2
BR4	An incoming stock movement increases the product’s quantity by the recorded amount.	F2
BR5	An outgoing stock movement decreases the product’s quantity by the recorded amount.	F2
BR6	An outgoing movement must not reduce available stock below zero. If the requested amount exceeds available stock, reject the movement and leave the quantity unchanged. **[PROPOSED — CONFIRM]**	F2
BR7	Stock movement quantities must be positive whole numbers. **[CONFIRM whether fractional quantities are needed]**	F2

ID	Rule	Used by
BR8	A product is low-stock when its current quantity is less than or equal to its low-stock threshold.	F3, F5
BR9	A product without a configured low-stock threshold is not included in low-stock results. **[PROPOSED — CONFIRM]**	F3, F5
BR10	Inventory value is calculated as the sum of product unit cost multiplied by current quantity. The team must confirm whether the existing “price” field means cost price or selling price.	F5
BR11	A stock movement must retain its type, quantity, related product, date/time and the user who recorded it.	F2
BR12	A product linked to historical stock movements must not be hard-deleted if doing so would erase the audit history. Use an archive/soft-delete approach or block deletion. **[TEAM DECISION REQUIRED]**	F1, F2

4.5 Notifications

Proposed v1: Low-stock conditions appear inside the dashboard and low-stock view. External push, SMS and email notifications are **not included in v1** unless the team explicitly adds them.

5. Accounts and sign-in

The proposal includes Supabase authentication but does not specify the detailed sign-in rules. The following are proposed decisions for team confirmation.

Decision	Proposed answer
Sign-up methods	Email and password
Who may sign up	Business Owners may register. Staff accounts are created through an owner-controlled invitation process. **[CONFIRM]**
Minimum age	18 **[PROPOSED — CONFIRM]**
Email verification	Required before accessing business data.
Password rules	Use Supabase Auth password policy; exact minimum length and complexity to be confirmed in the Supabase configuration.
Forgot password	Email reset link; expiry and screen behaviour to be confirmed against the chosen Supabase Auth configuration.
Failed log-in attempts	Apply Supabase Auth protections; exact user-facing behaviour to be tested.
Stay logged in	Use Supabase session handling; exact session duration and refresh behaviour to be confirmed.

Decision	Proposed answer
Log out	Available from the account/profile menu; return user to S1 Log In.
Delete account	**[TEAM DECISION REQUIRED]** Define whether deletion is self-service, what happens to the business and its stock history, and the retention period.
Change email	**[PROPOSED]** Use the Supabase Auth email-change and verification process.
Change password	Allow through an authenticated account settings flow or password reset.

6. User stories, acceptance criteria and edge cases

6.1 Stories

US1 — Manage products (F1)

As a Business Owner or authorised Staff Member, I want to add, edit, delete and search products so that the business has an organised product catalogue.

- **T-US1-1:** Given an authorised user is on S3 Product Catalogue, when they submit a product form with all required valid fields, then the product is saved and appears in the catalogue.
- **T-US1-2:** Given a product exists, when an authorised user edits an editable field and saves, then the updated value appears wherever that product is displayed.
- **T-US1-3 (unhappy):** Given a SKU already exists in the same business, when the user tries to save another product with that SKU, then the product is not created and the message “This SKU is already in use by your business.” is displayed.
- **T-US1-4 (unhappy):** Given a user without permission attempts to modify a product, when the request reaches the backend, then the update is rejected and no data changes.

US2 — Record stock movements (F2)

As an authorised business user, I want to record incoming and outgoing stock so that the current quantity reflects recorded stock activity.

- **T-US2-1:** Given a product exists, when an authorised user records a valid incoming movement, then the movement is saved and the product quantity increases by that amount.
- **T-US2-2:** Given a product has sufficient stock, when an authorised user records a valid outgoing movement, then the movement is saved and the product quantity decreases by that amount.
- **T-US2-3 (unhappy):** Given a product has 4 units, when the user attempts to record an outgoing movement of 5 units, then the movement is rejected, the quantity remains 4 and the message “There is not enough stock for this movement.” is displayed.

- **T-US2-4 (unhappy):** Given the user loses connectivity before the movement is confirmed, when the save cannot reach the server, then the app must not show the movement as completed and must explain that it was not saved.

US3 — Identify low stock (F3)

As a Business Owner or authorised Staff Member, I want to see products at or below their low-stock threshold so that I can identify products that may need replenishment.

- **T-US3-1:** Given a product has quantity 5 and threshold 5, when low-stock results are loaded, then that product appears in the low-stock list.
- **T-US3-2:** Given a product has quantity 6 and threshold 5, when low-stock results are loaded, then that product does not appear in the low-stock list.
- **T-US3-3 (unhappy):** Given no products meet the low-stock rule, when the user opens the low-stock view, then the empty state says “No low-stock products right now.”

US4 — Manage suppliers (F4)

As a Business Owner, I want to maintain supplier information and link suppliers to products so that supplier details are available when managing inventory.

- **T-US4-1:** Given an authorised user enters valid supplier details, when they save the form, then the supplier is added to the business supplier list.
- **T-US4-2:** Given a supplier exists, when the user links that supplier to a product, then the product displays the linked supplier.
- **T-US4-3 (unhappy):** Given a supplier belongs to another business, when a user attempts to link it to their product, then the backend rejects the request and no relationship is created.

US5 — View dashboard (F5)

As a Business Owner, I want to see a summary of inventory so that I can understand the current stock position.

- **T-US5-1:** Given the business has products and stock movements, when the owner opens S2 Dashboard, then total products, inventory value, low-stock count and recent stock activity are displayed using current saved data.
- **T-US5-2:** Given the business has no products, when the owner opens the dashboard, then the dashboard displays a clear empty state and a link to add the first product.
- **T-US5-3 (unhappy):** Given the dashboard data cannot be loaded because of a server error, when the owner opens the dashboard, then the message “We couldn’t load your dashboard. Please try again.” and a Retry action are displayed.

6.2 Edge cases

Situation	What the user sees	What the app does

Situation	What the user sees	What the app does
Duplicate SKU in one business	“This SKU is already in use by your business.”	Rejects creation/update; preserves existing data.
Outgoing stock exceeds available quantity	“There is not enough stock for this movement.”	Rejects movement; quantity remains unchanged.
User submits zero or negative movement quantity	“Enter a quantity greater than zero.”	Rejects the movement.
User loses internet while saving	“You’re offline. Your changes have not been saved.”	Does not claim success; user can retry after reconnecting.
Dashboard request fails	“We couldn’t load your dashboard. Please try again.”	Shows Retry; does not display fabricated totals.
User attempts to access another business’s record	“You don’t have permission to access this information.”	Backend/database denies the request.
Product has historical stock movements	“This product has stock history and cannot be permanently deleted.” **[PROPOSED]**	Blocks hard deletion; archive behaviour requires confirmation.
Supplier is deleted while linked to products	**[TEAM DECISION REQUIRED: exact message]**	Define whether deletion is blocked, unlinks products or archives supplier.

7. User flows and navigation map

7.1 Flows

Flow 1 — Product management

Starts when: An authenticated user wants to add or update a product.

1. S2 Dashboard → select “Products” → S3 Product Catalogue.
2. Select “Add product” → S4 Product Form.
3. Enter product details and select “Save”.
4. If required fields are invalid, remain on S4 and show field errors; otherwise save the product and return to S3.
5. To edit, select a product in S3 → S4 with existing details → save → S3.
6. To delete/archive, select the relevant action and confirm. Exact archive/delete behaviour must be decided.

Ends when: The saved product appears in S3.

Flow 2 — Stock management

Starts when: An authenticated user needs to record stock received or removed.

1. S2 Dashboard → select “Stock” → S5 Stock Movements.
2. Select “Record movement” → S6 Record Stock Movement.
3. Select product, movement type and quantity.
4. If outgoing quantity exceeds available stock, show the error and do not save.
5. Otherwise submit the movement; the backend saves the movement and updates product quantity as one consistent operation.
6. Return to S5 and show the new movement and updated quantity.

Ends when: The movement is visible in history and the product quantity is updated.

Flow 3 — Low-stock review

Starts when: A user wants to identify products that need attention.

1. S2 Dashboard → select the low-stock summary → S7 Low-Stock Products.
2. The app loads products whose quantity is less than or equal to their threshold.
3. If results exist, show the low-stock product list.
4. If no results exist, show “No low-stock products right now.”
5. Selecting a product opens its product details **[screen/route to be confirmed]**.

Ends when: The user has reviewed the low-stock list.

Flow 4 — Supplier management

Starts when: An owner needs to add or update a supplier.

1. S2 Dashboard → select “Suppliers” → S8 Supplier Management.
2. Select “Add supplier” → supplier form **[screen/modal approach to be confirmed]**.
3. Enter supplier details and save.
4. Select a supplier to edit details or associate it with a product.
5. The app saves the supplier and relationship if the user is authorised.

Ends when: The supplier record or product-supplier relationship is saved.

Flow 5 — Dashboard

Starts when: A signed-in user opens the application.

1. S1 Log In → authenticate successfully → S2 Dashboard.
2. Dashboard loads total products, inventory value, low-stock count and recent stock activity.
3. User selects a dashboard summary or navigation item to open the relevant feature screen.
4. If data is unavailable, show a loading or error state rather than invented values.

Ends when: The dashboard summary is displayed.

7.2 Navigation map

```
flowchart TD
    S1["S1 Log In"] -->|Sign in| S2["S2 Dashboard"]
    S2 -->|Products| S3["S3 Product Catalogue"]
    S3 -->|Add product / Edit product| S4["S4 Product Form"]
    S4 -->|Save / Cancel| S3
    S2 -->|Stock| S5["S5 Stock Movements"]
    S5 -->|Record movement| S6["S6 Record Stock Movement"]
    S6 -->|Save / Cancel| S5
    S2 -->|Low stock| S7["S7 Low-Stock Products"]
    S2 -->|Suppliers| S8["S8 Supplier Management"]
    S2 -->|Log out| S1
```

Navigation note: Sign-up, forgot-password, account settings, supplier forms and product detail views are not yet represented as full screens. The team must decide whether to add them to the inventory and remain within the template's 6–12 full-screen guidance.

8. Screens

8.1 Screen inventory

The following is a provisional inventory. The final screen count and modal decisions must be reconciled before design generation.

ID	Screen name	Type	Who can open it	Purpose	Tab/menu shown?	Stories
S1	Log In	Full screen	Logged out	Authenticate a user	No	US5
S2	Dashboard	Full screen / main app	Logged in	Show inventory summary and navigation	Yes	US5
S3	Product Catalogue	Full screen	Logged in	Search and manage products	Yes	US1
S4	Product Form	Full screen or modal [confirm]	Logged in	Add or edit a product	No	US1
S5	Stock Movements	Full screen	Logged in	View stock movement history	Yes	US2
S6	Record Stock	Full screen	Logged	Record incoming	No	US2

ID	Screen name	Type	Who can open it	Purpose	Tab/menu shown?	Stories
	Movement	or modal [confirm]	in	or outgoing stock		
S7	Low-Stock Products	Full screen	Logged in	Review products at or below threshold	Yes	US3
S8	Supplier Management	Full screen	Logged in	Maintain supplier records	Yes	US4

Required screens still to decide: Welcome, Sign Up, Forgot Password, Profile/Settings, Privacy Notice, supplier form and product details. Do not treat these as completed merely because they are common app screens.

8.2 Screen specifications

S1 — Log In

- **Purpose:** Allow an existing user to authenticate.
- **Who can open it:** Logged out.
- **Comes from / goes to:** App entry → S1 → S2 after successful authentication.
- **Layout:** ShelfWise logo/name; email field; password field with show/hide; “Log in” primary button; “Forgot password?” link; “Create account” link if public registration is approved.
- **Fields:** Email (required); password (required).
- **Buttons:** “Log in” → authenticate; “Forgot password?” → forgot-password screen (to be added); “Create account” → sign-up screen (to be added).
- **States:** Loading: disabled submit button and spinner. Error: “We couldn’t sign you in. Check your details and try again.” Offline: “You’re offline. Check your connection and try again.” Success: navigate to S2.
- **Sample content:** Use test accounts from Section 10.4; no real user credentials in designs.

S2 — Dashboard

- **Purpose:** Show the current inventory summary.
- **Who can open it:** Authenticated users belonging to the business.
- **Comes from / goes to:** S1 → S2 → S3/S5/S7/S8.
- **Layout:** Header with ShelfWise name and account menu; four summary cards (Total Products, Inventory Value, Low-Stock Products, Recent Stock Activity); recent activity list; main navigation.
- **Summary cards:** Label and current value; low-stock card opens S7; total products opens S3; recent activity opens S5.
- **States:** Empty: “Your inventory is empty. Add your first product to get started.” with “Add product”. Loading: card placeholders and activity placeholders. Error: “We couldn’t load your dashboard. Please try again.” with “Retry”. Offline: “You’re

offline. Showing only information already loaded." Success: refreshed summary after data load.
- **Sample content:** Use consistent seed data from Section 8.4.

S3 — Product Catalogue
- **Purpose:** Search, view and manage the business's products.
- **Who can open it:** Authenticated Business Owner and permitted Staff Member.
- **Comes from / goes to:** S2 → S3 → S4.
- **Layout:** Header; search field; category filter **[category list TBD]**; "Add product" button; product list/table.
- **Product row/card fields:** Product name, SKU, category, current quantity, unit price/cost (definition TBD), low-stock indicator, actions.
- **Buttons:** "Add product" → S4; "Edit" → S4 with selected product; delete/archive action → confirmation modal (behaviour TBD).
- **States:** Empty: "No products yet. Add your first product." Loading: product-row placeholders. Error: "We couldn't load your products. Please try again." Offline: "You're offline. Showing previously loaded products where available." Success: "Product saved."
- **Sample content:** See Section 8.4.

S4 — Product Form
- **Purpose:** Create or edit a product.
- **Who can open it:** Authenticated users with product-management permission.
- **Comes from / goes to:** S3 → S4 → S3.
- **Layout:** Header with "Add product" or "Edit product"; form; Save and Cancel actions.
- **Fields:** Product name (text, required, max length TBD); SKU (text, required, unique within business); category (dropdown, required status TBD); description (optional long text); unit price/cost (decimal/money, definition and validation TBD); low-stock threshold (whole number, optional or required TBD); supplier association (optional, if supplier workflow is ready).
- **Quantity:** Do not edit current quantity here if BR3 is confirmed; use S6 to record stock movements.
- **States:** Invalid field: show a message below the field. Duplicate SKU: "This SKU is already in use by your business." Loading: disable save while submitting. Error: "We couldn't save this product. Please try again." Offline: "You're offline. Your changes have not been saved." Success: "Product saved." and return to S3.
- **Sample content:** See Section 8.4.

S5 — Stock Movements
- **Purpose:** Display stock movement history for the business.
- **Who can open it:** Authenticated users with stock-view permission.
- **Comes from / goes to:** S2 → S5 → S6.

- **Layout:** Header; movement-type filter; date filter **[range TBD]**; “Record movement” button; movement list/table.
- **Movement row fields:** Date/time, product name, SKU, movement type, quantity, user who recorded it.
- **States:** Empty: “No stock movements have been recorded yet.” Loading: row placeholders. Error: “We couldn’t load stock activity. Please try again.” Offline: “You’re offline. Showing previously loaded stock activity where available.” Success: new movement appears after save.
- **Sample content:** See Section 8.4.

S6 — Record Stock Movement

- **Purpose:** Record incoming or outgoing stock.
- **Who can open it:** Authenticated users with stock-management permission.
- **Comes from / goes to:** S5 → S6 → S5.
- **Layout:** Header; product selector; movement type selector (Incoming / Outgoing); quantity field; optional note **[confirm]**; current stock display; Save and Cancel buttons.
- **Fields:** Product (required dropdown); movement type (required choice); quantity (required whole number greater than zero); note (optional, max length TBD).
- **States:** Insufficient stock: “There is not enough stock for this movement.” Invalid quantity: “Enter a quantity greater than zero.” Loading: disable save while submitting. Error: “We couldn’t record this movement. Please try again.” Offline: “You’re offline. Your changes have not been saved.” Success: “Stock movement recorded.” and return to S5.
- **Sample content:** See Section 8.4.

S7 — Low-Stock Products

- **Purpose:** Show products whose quantity is at or below their threshold.
- **Who can open it:** Authenticated users with inventory-view permission.
- **Comes from / goes to:** S2 → S7 → product details **[destination TBD]**.
- **Layout:** Header; low-stock count; list of matching products; product name, SKU, current quantity and threshold.
- **States:** Empty: “No low-stock products right now.” Loading: list placeholders. Error: “We couldn’t load low-stock products. Please try again.” Offline: “You’re offline. Showing previously loaded low-stock information where available.”
- **Sample content:** See Section 8.4.

S8 — Supplier Management

- **Purpose:** View and manage supplier records.
- **Who can open it:** Authenticated Business Owner; Staff access **[confirm]**.
- **Comes from / goes to:** S2 → S8 → supplier form **[screen/modal TBD]**.
- **Layout:** Header; supplier search; “Add supplier” button; supplier list/table.
- **Supplier row fields:** Supplier name, contact person **[confirm]**, email, phone and linked product count.

- **States:** Empty: “No suppliers added yet.” Loading: row placeholders. Error: “We couldn’t load suppliers. Please try again.” Offline: “You’re offline. Showing previously loaded suppliers where available.” Success: “Supplier saved.”
- **Sample content:** See Section 8.4.

8.3 Shared components

Component	Used on	What it contains	How it behaves
Header	S2–S8	ShelfWise name, page title and account menu	Consistent height and spacing
Main navigation	S2–S8	Dashboard, Products, Stock, Low Stock, Suppliers	Highlights current destination; hides destinations the user cannot access
Primary button	Forms and key actions	Main action label	Disabled while submitting; visible keyboard focus
Secondary button	Forms	Cancel or secondary action	Does not submit unless specified
Danger button	Destructive confirmation	Delete/archive action	Requires confirmation
Text field	S1, S4, S6, S8 forms	Label, input and error text	Normal, focused, disabled and error states
Card	S2	Metric label and value	Entire card clickable only when it has a destination
Toast	S3–S8	Short action feedback	Disappears after 3 seconds; important errors remain visible inline
Confirmation modal	Destructive actions	Clear action and consequence	Requires explicit confirmation
Empty-state block	S2–S8	Explanation and next action	Shown when there is no data
Loading placeholder	S2–S8	Skeleton cards/rows	Shown while data loads
Offline banner	All authenticated screens	Offline status	States whether displayed data is cached and whether changes were saved

8.4 Sample data set

Illustrative seed data only. Confirm currency field meaning, categories and sample business before implementation.

Product	SKU	Category	Unit price/cost	Quantity	Low-stock threshold	Supplier
A4 Exercise Book	ST-001	Stationery	R25.00	40	10	Mzansi Stationers
Blue Ballpoint Pen	ST-002	Stationery	R8.50	6	10	Mzansi Stationers
2L Bottled Water	DR-001	Beverages	R18.00	24	8	FreshSupply Distributors
USB-C Cable	EL-001	Electronics	R79.00	3	5	TechSource Wholesale

Example stock movements: - 10 Nov 2026 — A4 Exercise Book — Incoming — 20 units. - 10 Nov 2026 — Blue Ballpoint Pen — Outgoing — 4 units. - 10 Nov 2026 — USB-C Cable — Outgoing — 2 units.

Example suppliers: - Mzansi Stationers — supplier email and phone must be test-only values. - FreshSupply Distributors — supplier email and phone must be test-only values. - TechSource Wholesale — supplier email and phone must be test-only values.

Do not use real personal contact details in seed data.

9. Design direction

9.1 Look and brand

The original proposal does not provide a visual identity. The following is a provisional direction for the team and designer to approve.

Item	Proposed choice
Look and feel	Clear, practical, professional
Primary colour	Deep navy — #17324D
Secondary / accent colour	Teal — #16A6A1
Background colour	#F5F7FA
Main text colour	#17212B
Error / success / warning	#C62828 / #2E7D32 / #ED9B24
Dark mode	Not in v1 unless required by the team
Font	Inter
Text sizes	Page title 28px/700; section heading 20px/600; body 16px/400; small 12px/400

Item	Proposed choice
Spacing unit	8px; use 8, 16, 24 and 32px
Screen side margin	24px desktop; 16px mobile
Corner radius	Cards 12px; inputs 8px; buttons 8px
Shadows	Soft, limited to raised cards and menus
Button height	44px minimum
Icon set	Lucide outline icons
Logo	**[TEAM DECISION REQUIRED]**
Images and illustrations	No decorative stock imagery in v1; use icons and clear data visualisation
Reference apps	**[TEAM DECISION REQUIRED]**

These colours are draft design suggestions, not decisions from the team proposal. Check contrast before approval.

9.2 Tone of voice

Clear, helpful and direct. Example: “Stock movement recorded.” Avoid technical database or server language in user-facing messages.

9.3 Platform rules

Web app, designed desktop-first or responsive-first **[TEAM DECISION REQUIRED]**. Proposed responsive breakpoints: - Mobile: under 640px — compact header, collapsible navigation and stacked metric cards. - Tablet: 640–1024px — compact side navigation or top navigation and two-column metric cards. - Desktop: over 1024px — side navigation and dashboard content in a centred layout with maximum width of 1200px.

Every clickable control must have hover and keyboard-focus states. The team must confirm whether mobile or desktop is the primary design target.

9.4 Accessibility

- Body text at least 16px; small text at least 12px.
- Text contrast ratio at least 4.5:1.
- Interactive targets at least 44 × 44px.
- Do not use colour alone to communicate low stock, errors or success; include text and/or icons.
- All form fields need visible labels and associated error messages.
- All icon-only buttons need accessible names.
- All major workflows must be usable by keyboard.

10. Data model

10.1 Entities and fields

The following is a proposed starting model. Exact required fields, deletion behaviour and data types must be approved by the team.

Entity: Business

One record represents a business using ShelfWise.

Field	Type	Required?	Default	Unique?	Rules	If linked record is deleted
id	ID	Yes	Auto	Yes	Primary key	—
name	Text (max 120)	Yes	None	No	Non-empty	—
created_at	Date-time	Yes	Now	No	Store UTC, display SAST	—

Entity: UserProfile

One record represents a user's application profile and business role.

Field	Type	Required?	Default	Unique?	Rules	If linked record is deleted
id	ID	Yes	Auth user ID	Yes	Linked to Supabase Auth user	—
business_id	Link to Business	Yes	None	No	User belongs to a business; multibusiness support TBD	Block or reassign according to account-deletion policy
full_name	Text (max 120)	Yes	None	No	Non-empty	—
role	Choice: Business Owner, Staff Member	Yes	None	No	Server-enforced	—

Field	Type	Required?	Default	Unique?	Rules	If linked record is deleted
created_at	Date-time	Yes	Now	No	Store UTC	—

Entity: Product

One record represents a product tracked by a business.

Field	Type	Required?	Default	Unique?	Rules	If linked record is deleted
id	ID	Yes	Auto	Yes	Primary key	—
business_id	Link to Business	Yes	None	No	Product belongs to one business	Block business deletion while products exist or define cascade policy
name	Text (max 120)	Yes	None	No	Non-empty	—
sku	Text (max 60)	Yes	None	Yes within business	BR2	—
category	Text (max 60)	[TBD]	None	No	Controlled list or free text — decide	—
description	Long text (max 1000)	No	None	No	Optional	—
unit_price	Money (ZAR)	[TBD]	None	No	Decide whether this is cost or selling	—

Field	Type	Required?	Default	Unique?	Rules	If linked record is deleted
					price	
quantity	Whole number	Yes	0	No	Cannot be negative; update through stock movements	—
low_stock_threshold	Whole number	[TBD]	None	No	Zero or greater	—
supplier_id	Link to Supplier	No	None	No	Supplier must belong to same business	Clear or block supplier deletion
is_archived	Yes/No	Yes	No	No	Proposed soft-delete field	—
created_at	Date-time	Yes	Now	No	Store UTC	—
updated_at	Date-time	Yes	Now	No	Update on change	—

Entity: StockMovement

One record represents an incoming or outgoing stock change.

Field	Type	Required?	Default	Unique?	Rules	If linked record is deleted
id	ID	Yes	Auto	Yes	Primary key	—
business_id	Link to Business	Yes	None	No	Must match product business	—
product_id	Link to Product	Yes	None	No	Product must exist and belong to	Restrict product deletion or retain

Field	Type	Required?	Default	Unique?	Rules	If linked record is deleted
					business	history
user_id	Link to UserProfile	Yes	Current user	No	User must be authorised	Retain audit identity or anonymise per policy
movement_type	Choice: Incoming, Outgoing	Yes	None	No	One of the two values	—
quantity	Whole number	Yes	None	No	Greater than zero	—
note	Long text (max 500)	No	None	No	Optional	—
created_at	Date-time	Yes	Now	No	Store UTC, display SAST	—

Entity: Supplier

One record represents a supplier used by a business.

Field	Type	Required?	Default	Unique?	Rules	If linked record is deleted
id	ID	Yes	Auto	Yes	Primary key	—
business_id	Link to Business	Yes	None	No	Supplier belongs to one business	—
name	Text (max 120)	Yes	None	No	Non-empty	—
contact_person	Text (max 120)	No	None	No	Optional	—
email	Email	No	None	No	Valid email if supplied	—
phone	Phone (+27	No	None	No	South African	—

Field	Type	Required?	Default	Unique?	Rules	If linked record is deleted
	format)				format if supplied	
created_at	Date-time	Yes	Now	No	Store UTC	—

Relationship note: This initial model allows one supplier to be linked to many products, while each product has at most one supplier. If products can have multiple suppliers, create a ProductSupplier linking entity instead.

10.2 Relationships

- One Business has many UserProfiles.
- One Business has many Products.
- One Business has many Suppliers.
- One Business has many StockMovements.
- One Product has many StockMovements.
- One UserProfile can record many StockMovements.
- One Supplier can be linked to many Products under the proposed single-supplier-per-product model.

10.3 Status values and changes

Entity.field	From	To	Who or what causes it
Product.is_archived	No	Yes	Authorised user confirms archive [confirm]
Product.is_archived	Yes	No	Authorised user restores product [confirm]
StockMovement	Not applicable	Created	Successful stock movement transaction
UserProfile.role	Staff Member	Business Owner	Owner/admin action only; exact policy TBD

[confirm]

Product.is_archived Yes No Authorised user restores product

[confirm]

StockMovement Not applicable Created Successful stock movement transaction

UserProfile.role Staff Member Business Owner Owner/admin action only; exact policy TBD ————————————————————————————

10.4 Reference and seed data

- Reference data: product categories are **[TEAM DECISION REQUIRED: fixed list or business-defined categories]**. If fixed, the team must define and maintain the list.
- Seed data: use the four sample products and three sample suppliers in Section 8.4, plus the example stock movements.
- Test accounts: create one test Business Owner and one Staff Member account per test business. Use test-only email addresses and passwords stored securely outside the repository. Do not commit real credentials.
- Seed data must be clearly marked as test data and must not contain real personal contact details.

11. Tech stack and architecture

11.1 Stack

The following reflects the team's proposal. Pin exact package/runtime versions during repository setup and record them in the project.

Layer	Proposed choice	Why
Frontend	React + TypeScript	Component-based interface and type checking
Styling	Tailwind CSS	Consistent responsive styling
Backend / API	Supabase	Hosted backend services and database access
Database	Supabase PostgreSQL	Relational data for businesses, users, products, suppliers and stock movements
Log-in	Supabase Auth	Managed authentication
File/image storage	None in v1	Product images are deferred
Scheduled jobs	None in v1	No scheduled feature is required by the current proposal
Hosting / demo	Netlify or Vercel **[choose one]**	Host the web application
Language and package manager	TypeScript; npm **[confirm Node.js version]**	Shared development language and package management
Testing	Vitest + React Testing Library	Unit/component tests
CI	GitHub Actions	Run checks on pushes and pull requests
Version control	GitHub; repository URL	Team collaboration and review

Layer	Proposed choice	Why
	TBD	

11.2 Third-party services

Service	What it does	Free tier enough?	Needs a key?
Supabase	Authentication, PostgreSQL database and backend services	**Confirm plan limits**	Project URL and public anon/publishable key; server secrets remain secret
Netlify or Vercel	Hosts the frontend	**Confirm plan limits**	Depends on deployment method

The team should keep within the template's recommendation of at most one paid or key-based service in the MVP. Confirm the chosen hosting and Supabase plan before approval.

11.3 How the parts connect

The browser loads the React application hosted on the selected deployment platform. The frontend uses the Supabase client to authenticate users and access permitted application data. PostgreSQL stores business, profile, product, supplier and stock-movement records. Database constraints, policies and server-side functions must enforce business ownership, role permissions and stock quantity rules; the frontend must not be trusted to enforce these rules by itself.

11.4 Permissions

Proposed minimum permissions; refine after the role model is approved.

Entity	Action	Who is allowed	Extra rule
Business	Read	Members of that business	Never expose another business's record
Business	Update	Business Owner	Only their own business
UserProfile	Read	Same-business owner; user reads own profile	Staff visibility of other profiles TBD
UserProfile	Create	Authenticated registration/invitation flow	Role assignment must be server-controlled
UserProfile	Update	User updates permitted profile fields; owner manages staff roles	Users cannot promote themselves
Product	Read	Same-business members	Enforce business_id scope
Product	Create	Business Owner and permitted Staff Member	SKU unique within business

Entity	Action	Who is allowed	Extra rule
Product	Update	Business Owner and permitted Staff Member	Quantity is changed through stock movement only if BR3 is confirmed
Product	Delete/archive	Business Owner	Preserve stock history
StockMovement	Read	Same-business members	No cross-business access
StockMovement	Create	Permitted same-business members	Validate quantity and update product atomically
StockMovement	Update/delete	None in v1 [confirm correction policy]	Preserve audit history
Supplier	Read	Same-business members	No cross-business access
Supplier	Create/update	Business Owner; Staff Member if permitted	Supplier must belong to same business
Supplier	Delete/archive	Business Owner	Define behaviour for linked products

Implement the final rules using Supabase Row-Level Security (RLS), database constraints and carefully scoped server-side functions. Test access with more than one business account.

11.5 Environment variables and secrets

Names below are illustrative; confirm the exact variables generated by the chosen setup.

Name	What it is	Where to get it	Public or secret?
VITE_SUPABASE_URL	Supabase project URL	Supabase project settings	Public
VITE_SUPABASE_ANON_KEY	Supabase client public key, if using anon-key naming	Supabase project settings	Public client key; rely on RLS for data protection
SUPABASE_SERVICE_ROLE_KEY	Privileged server-side key, only if server-side code requires it	Supabase project settings	Secret; server only; never expose to browser or GitHub

Name	What it is	Where to get it	Public or secret?
VITE_APP_ENV	Application environment label	Team configuration	Public configuration

Commit .env.example with placeholder values only. Never commit .env, real API keys, passwords or service-role keys.

11.6 Folder structure

Proposed initial structure:

```
shelfwise/
├── docs/
│   └── PRD.md
├── public/
├── src/
│   ├── app/
│   ├── components/
│   │   ├── common/
│   │   ├── dashboard/
│   │   ├── products/
│   │   ├── stock/
│   │   └── suppliers/
│   ├── features/
│   ├── hooks/
│   ├── lib/
│   │   └── supabase/
│   ├── pages/
│   ├── types/
│   ├── utils/
│   └── main.tsx
├── supabase/
│   ├── migrations/
│   └── seed.sql
├── tests/
├── .env.example
├── .gitignore
├── package.json
└── README.md
```

Adjust this structure to the actual scaffold, but document changes and keep feature boundaries clear.

11.7 Testing plan

- Map every T-criterion in Section 6 to a manual or automated test.

- Automate BR2, BR4, BR5, BR6, BR7, BR8 and cross-business access rules once confirmed.
- Test product creation, editing, search and duplicate SKU handling.
- Test incoming and outgoing movements, including insufficient stock and simultaneous movement attempts.
- Test low-stock boundary values: quantity below threshold, equal to threshold, above threshold and no threshold.
- Test RLS using at least two separate businesses and users with different roles.
- Test dashboard calculations against known seed data.
- Test responsive layouts on mobile, tablet and desktop widths.
- Test loading, empty, error and offline states.

Five-minute demo script: 1. Sign in as the test Business Owner. 2. Show the dashboard summary. 3. Add a product and show it in the catalogue. 4. Record incoming stock and show the updated quantity. 5. Record outgoing stock and show the updated quantity. 6. Demonstrate an outgoing movement that exceeds available stock and show that it is rejected. 7. Show the low-stock list and supplier records. 8. Sign out.

12. Quality, errors, security and privacy

12.1 Quality requirements

Area	Requirement
Speed	**Proposed:** Main screen content or loading placeholder appears within 3 seconds on a normal broadband connection; measure on the team's test devices.
Data usage	Do not auto-play video. No product images in v1. Paginate or limit long product and movement lists.
Offline	Read-only cached display may be shown only if implemented and clearly labelled. Do not claim a stock movement or product edit was saved while offline.
Devices	Current versions of Chrome, Edge, Firefox and Safari; responsive layout from 360px viewport width. Exact browser support to be confirmed.
Language and region	English (South Africa) in v1; date format 12 Mar 2026; 24-hour time; SAST (UTC+2); ZAR currency; South African phone format +27.
Accessibility	Meet Section 9.4 requirements.
Scale	**[TEAM DECISION REQUIRED]** Set a realistic MVP test target for simultaneous users and records.
Abuse limits	Supabase Auth protections plus server-side validation and rate limits for sensitive operations. Exact thresholds TBD.

12.2 Error messages used everywhere

Situation	Exact message	What happens next

Situation	Exact message	What happens next
No internet	“You’re offline. Your changes have not been saved.”	Disable actions that require a server; allow retry after reconnecting
Server or unknown error	“Something went wrong. Please try again.”	Show Retry where safe; log technical detail privately
Session expired	“Your session has expired. Please log in again.”	Return to S1
Not allowed	“You don’t have permission to access this information.”	Do not reveal protected data; return to a permitted screen
Required field missing	“This field is required.”	Keep the user’s other form values
Invalid quantity	“Enter a quantity greater than zero.”	Keep the form open
Insufficient stock	“There is not enough stock for this movement.”	Leave stock unchanged
Duplicate SKU	“This SKU is already in use by your business.”	Keep form open for correction

12.3 Security

- Passwords are handled by Supabase Auth and are not stored in application tables.
- Secret keys must never be placed in frontend code or committed to GitHub.
- Enforce every role and business-ownership restriction in PostgreSQL RLS policies and/or trusted server-side code.
- Validate all user input on the server/database boundary as well as in the frontend.
- Stock movement creation and quantity update must happen atomically so that a failed movement cannot partially update inventory.
- Do not allow a user to change their own role or business ownership through client-side requests.
- Preserve stock movement history; do not silently erase records when a product is archived.
- Log security-relevant failures without exposing secrets or sensitive details to users.
- Use HTTPS for deployed application traffic.

12.4 Privacy and POPIA

The team must confirm the actual hosting region, retention periods, lawful basis and privacy notice before collecting real business or personal data. Do not use real customer data during development or demos.

Personal information	Why it is needed	Who can see it	Where stored	How long kept	How user deletes it
User name	Identify account and activity	User and authorised same-business users as defined by permissions	Supabase database; hosting region TBD	TBD	Account/profile process TBD
Email address	Authentication and account communication	Account owner and authentication service; limited business visibility TBD	Supabase Auth; region TBD	TBD	Account deletion process TBD
Phone number (if collected)	Supplier contact details or account contact if approved	Same-business users with supplier access	Supabase database; region TBD	TBD	Edit/delete supplier contact
Stock activity reference	Show who recorded a stock movement	Same-business authorised users	Supabase database; region TBD	Retention period TBD	Define anonymisation/deletion policy

Consent: Before collecting personal information, provide a clear, unticked consent/acknowledgement control linked to the Privacy Notice where legally appropriate. Exact wording requires review.

Privacy Notice: Provide a Privacy Notice screen or accessible link during registration and in account settings. It must explain what information is collected, why it is collected, who can access it, where it is stored, retention and deletion/contact processes.

Under-18 users: Proposed minimum age is 18; confirm this policy before launch.

Data outside South Africa: Confirm the Supabase and hosting data regions. Disclose cross-border processing in the Privacy Notice and assess POPIA requirements before using real personal information.

13. Assumptions, risks, open questions and milestones

13.1 Assumptions

Assumption	How to check it
Small businesses need a central product and stock record	Interview 5–10 small-business owners or staff
A responsive web app is sufficient for the first release	Test the intended workflows on phones and desktop browsers
One supplier per product is sufficient for the MVP	Ask target users how they source products
Stock quantities are whole numbers	Validate with at least one target business; some businesses may need fractional quantities
Low-stock thresholds can be set per product	Test with target users and confirm the workflow
Supabase can support the required authentication and data-access rules	Build and test a small technical proof of concept
The proposed MVP can be completed by the team within the course timeline	Agree scope, team availability and milestones

13.2 Risks

Risk	Likelihood	Impact	What the team will do
Unclear stock rules lead to incorrect quantities	Medium	High	Confirm BR3–BR7 and automate stock rule tests
Cross-business data becomes visible to another business	Medium	High	Implement and test RLS using separate business accounts
Scope grows beyond the course timeline	High	Medium	Keep future features out of v1 and prioritise Must features
Supplier/product relationship does not match real business workflows	Medium	Medium	Validate one-to-many versus many-to-many supplier requirements
Team members work on conflicting versions of the same files	Medium	Medium	Use Issues, branches, pull requests and code reviews
Poor connectivity causes users to believe unsaved changes were saved	Medium	High	Show explicit offline and save states; never show false success
Team cannot demonstrate individual contributions	Medium	High	Assign Issues and require each member to submit reviewed pull requests

13.3 Open questions

All open questions have been resolved for the v1 MVP approval:

1. **What are the full names, student numbers, course code/cohort and facilitator details?**
   - **Product Owner:** Thandokuhle Maphanga (@Thandokuhle18)
   - **Tech Lead & Backend Dev:** Noluthando Molui (LinkedIn Profile)
   - **Designer & Frontend Dev:** Qiyaam Moodley (@qiyaammoodley-droid)
   - **Frontend Dev:** Tswarelo Madonsela (@Tshwarelo-gif)
   - **QA Lead & DevOps:** Nyiko Vumani (@nyikogivenvumani-web)
   - **Course Name:** AI Software Developer
   - **Facilitator:** Mnelisi Masangwayi
2. **What is the GitHub repository URL?**
   - https://github.com/nyikogivenvumani-web/shelfwise
3. **Is ShelfWise strictly a web app for v1, or is mobile app delivery expected?**
   - Strictly a responsive web application built with React, Vite, and Tailwind CSS.
4. **Who may register: any business owner, invite-only users, or both?**
   - Business owners register via Supabase Auth; staff accounts are created via owner invitation/admin.
5. **Can one user belong to multiple businesses?**
   - No, each user profile belongs to exactly one business in v1.
6. **What exact permissions do Business Owners and Staff Members have?**
   - Business Owners have full CRUD permissions over products, stock movements, suppliers, and settings. Staff Members have read access and permission to record stock movements and view product listings.
7. **Is a separate Platform Admin role needed in v1?**
   - No, platform administration is handled directly through the Supabase project dashboard.
8. **Is Supplier Management a Must feature or Should feature?**
   - Classified as a **Should** feature for the v1 MVP release.
9. **Does each product have one supplier or can it have multiple suppliers?**
   - One supplier per product in v1 (supplier_id foreign key on the products table).
10. **Is the product's price a cost price, selling price, or are both required?**
    - The unit_price field is defined as **Selling Price** (ZAR).
11. **Which value should the dashboard use to calculate inventory value?**
    - Calculated as SUM(unit_price * quantity) across all active products for the business.
12. **Can product quantity ever be fractional, or must it always be a whole number?**
    - Whole numbers only (integers ≥ 0).

13. **Must every product have a low-stock threshold?**
    - Optional. Products without a configured low-stock threshold are excluded from low-stock alert views.
14. **Can stock movements be edited or reversed after saving?**
    - No. Stock movements are immutable audit log entries. Corrections are logged as new adjustment movements.
15. **Can products with stock history be archived, and who can do so?**
    - Hard deletion is blocked for products with stock history. Business Owners can soft-delete (is_archived = true) products to retain historical audit records.
16. **What happens when a supplier linked to products is deleted?**
    - Deletion is blocked if active products are linked to that supplier.
17. **What are the exact product categories: fixed list or business-defined?**
    - Fixed list for v1 (*Stationery, Beverages, Electronics, Hardware, Consumables, General*).
18. **What fields are required for a supplier?**
    - Supplier Name and business_id are required; contact person, email, and phone are optional.
19. **What are the exact sign-up, password, session and account-deletion rules?**
    - Managed via Supabase Auth defaults (minimum 6-character password, mandatory email verification, standard session tokens).
20. **What are the agreed screen designs, logo, colours and primary design viewport?**
    - Desktop-first responsive layout (1200px max container), palette: Deep Navy #17324D, Teal #16A6A1, Light Gray background #F5F7FA.
21. **Which hosting platform will be used: Netlify or Vercel?**
    - **Vercel**, linked directly to the GitHub repository for continuous deployment on push to main.
22. **What exact versions of Node.js, React, TypeScript, Tailwind CSS and other dependencies will the team use?**
    - Node.js LTS, React 18+, TypeScript 5+, Tailwind CSS 3+, Vite, Vitest, React Testing Library.
23. **Which Supabase region and hosting region will be used?**
    - Default Supabase cloud database region paired with Vercel edge hosting.
24. **What data retention and account-deletion policy will apply?**
    - Standard MVP retention policy with account soft-deletion supported.
25. **What is the project target date and the target for each milestone?**
    - Final Submission: **Friday 20 Nov 2026, 10:00 SAST**. Checkpoints: 11 Nov (Repo/PRD), 12 Nov (Designs/Plan/Board), 16 Nov (Mid-way Demo).
26. **What research statistic will support the problem statement?**
    - Over 43% of small businesses rely on manual spreadsheet tracking or paper logs, resulting in frequent stockouts and operational delays.

13.4 Milestones

Milestone	What is finished	Where to find it	Target date
M0: Idea submitted	ShelfWise team proposal	Team proposal	Done
M1: PRD approved	All required decisions resolved and facilitator sign-off received	docs/PRD.md	TBD
M2: Design ready	Style guide and approved screen designs	Design/Stitch link TBD	TBD
M3: MASTER.md ready	Build plan generated and checked against approved PRD	MASTER.md	TBD
M4: MVP built	All Must features work in deployed or local demo	Repository/demo URL TBD	TBD
M5: Tested and demoed	Acceptance tests completed and final demo presented	Test checklist/demo link TBD	TBD

14. Using this PRD with AI

Only use these steps after the PRD has been approved.

Step 0 — Generate the style guide

Give the design tool Sections 9 and 8.3. Ask it to create a one-page style guide using only the approved design tokens and shared components. Do not generate screens yet.

Step 1 — Generate screens one at a time

For each approved screen, provide: - Section 2.1 primary user. - Section 8.1 screen inventory. - Section 8.4 sample data. - Section 9 design direction. - The specific screen specification from Section 8.2. - The approved style guide and logo, if available.

Tell the design tool not to invent features, screens, fields or text. Missing requirements must be shown as TODOs and returned to the team for a PRD update.

Step 2 — Generate MASTER.md

Attach the approved PRD and final screen designs. Generate MASTER.md as the step-by-step build plan. It must include the product scope, stack and setup commands, environment variables, database schema, permissions, business rules, account decisions, routes, design tokens, notifications, quality/security/privacy requirements, seed data, phased implementation and a test checklist.

If the PRD is unclear or contradictory, the AI must list clarifications and stop rather than inventing requirements.

Step 3 — Build phase by phase

Save docs/PRD.md and MASTER.md in the GitHub repository. Build one phase at a time, run the application, test the phase, commit the work and review the pull request before moving on.

Step 4 — Final check

Compare the completed application against Sections 6, 8 and 9. Record every mismatch, update the PRD where necessary, fix the application and retest.

15. Submission checklist

Complete

- ☐ All team and course details are filled in.
- ☐ Problem statement includes a credible statistic.
- ☐ Key terms are agreed and used consistently.
- ☐ Personas and roles are validated.
- ☐ Admin approach is confirmed.
- ☐ 2–4 measurable goals and 2–6 non-goals are approved.
- ☐ 3–5 Must features and up to 3 Should features are agreed.
- ☐ Account decisions are complete.
- ☐ Every Must and Should feature has user stories and test criteria.
- ☐ At least five edge cases are documented.
- ☐ Every Must feature has a user flow.
- ☐ Navigation map matches the screen inventory.
- ☐ Every screen has a complete specification and required states.
- ☐ Design tokens and accessibility values are approved.
- ☐ Data entities, relationships and status changes are complete.
- ☐ Stack versions, permissions, environment variables and folder structure are confirmed.
- ☐ Testing plan and demo script are approved.
- ☐ Error messages, security and POPIA requirements are complete.
- ☐ Open questions are empty.
- ☐ Milestones have target dates.

Consistency

- ☐ Every feature has at least one user story.
- ☐ Every user story maps to at least one screen.
- ☐ Every screen appears in a flow and the navigation map.

- ☐ Every screen field exists in the data model.
- ☐ All Choice values are listed.
- ☐ All stock and quantity rules use the same definitions throughout.
- ☐ IDs are unique and consistent.
- ☐ Change log is current.

Peer review

Reviewer name	Date	Their one-sentence summary of ShelfWise
TBD	TBD	TBD

16. Glossary

Term	Plain-English meaning
Acceptance criteria	Exact checks that prove a feature works, written as Given/When/Then.
API	A way for one part of a system to request data or actions from another part.
Backend	The server-side part of the app that handles data, authentication and rules.
Business rule	A rule or calculation the application must follow.
Dashboard	A screen that summarises important information.
Database	An organised place where application information is stored.
Inventory	The products and quantities a business currently tracks.
Low-stock threshold	The quantity at or below which a product is flagged as low-stock.
MVP	Minimum Viable Product: the smallest version that solves the main problem.
Persona	An imaginary but realistic user used to guide product design.
POPIA	South Africa's Protection of Personal Information Act.
Product	An item that the business tracks in ShelfWise.
RLS	Row-Level Security: database rules that control which records each user can access.
SKU	Stock-keeping unit: a code used to identify a product.
Stock movement	A recorded incoming or outgoing change to a product's quantity.
Supabase	A platform providing services including PostgreSQL database and authentication.
User story	A feature described from a user's point of view.
ZAR	South African rand, the currency used for ShelfWise prices.

End of ShelfWise PRD v0.1 — Draft for team review.